

MEDIATEK

双系统安全启动远程签名

V0.1

Revision History

Version	Date	Author	Description
0.1	2026-01-28	Steven Gao	Initial draft

签名Image清单

bl2.img, vbmeta.img, vbmeta_system.img, vbmeta_vendor.img, modem.img, spmfw.img, pi_img.img, dpm.img, scp.img, vcp.img, sspm.img, mcupm.img, gpueb.img, apusys.img, gz.img, boot.img, vendor_boot.img, init_boot.img, dtbo.img, tee.img, consys_gnss.img, connsys_wifi.img, consys_bt.img, logo.img, audio_dsp.img, super.img, system.ext4,bl2-an.img, yocto-boot.img

未签名image清单

mcf_ota.img, ccu.img, userdata.ext4, userdata.img

1. 准备公钥和证书

从安全部门获取secure boot 使用的公钥，Android 和Yocto key 可以一样

Key Name	Key 类型	Key 用途
root_pubk.pem	Android root 公钥	Android prebuilt image 签名
img_pubk.pem	Android image 公钥	Android prebuilt image 签名
sbc_key.pem	yocto root 公钥	Yocto lk签名
verified_key.pem	yocto image 公钥	Yocto lk, fit image, rootfs签名
verified_key.crt	yocto image 证书	Yocto fit image 公钥

2. 替换本地环境的KEY—Android签名key

1. 抓取android branch
2. `cd vendor/mediatek/proprietary/scripts/sign-image_v2`
3. 将第4页root pubk key 和image pubk key 复制到hsm_test_keys目录下
4. `python img_key_deploy.py mt6881 cert1_key_path=hsm_test_keys
/root_pubk.pem cert2_key_path=hsm_test_keys/img_pubk.pem
root_key_padding=pss | tee gen_cert1_cert2_key.log`
5. 更新内容会在 `vendor/mediatek/proprietary/custom/mt6881/security/cert_config`

2. 替换本地环境的KEY—Android签名key

6. 将上一步的两把公钥继续更新到<yocto branch>/meta/meta-mediatek-mt8668/scripts/sign-image_v2/hsm_test_keys

meta > meta-mediatek-mt8668 > scripts > sign-image_v2 > hsm_test_keys			
Name	Date modified	Type	Size
 img_prvk.pem	2025/9/16 11:16	PEM File	2 KB
 img_pubk.pem	2025/9/16 11:16	PEM File	1 KB
 root_prvk.pem	2025/9/16 11:16	PEM File	2 KB
 root_pubk.pem	2025/9/16 11:16	PEM File	1 KB

7. 在上一级sign-image_v2目录执行以下命令生成签名用的证书,
python img_key_deploy.py mt6881
cert1_key_path=hsm_test_keys/root_pubk.pem
cert2_key_path=hsm_test_keys/img_pubk.pem root_key_padding=pss | tee
gen_cert1_cert2_key.log

2. 替换本地环境的KEY—Yocto签名key

将第4页Yocto 三把公钥更新到<yocto branch>/meta/meta-mediatek/conf/machine/keys/ 目录下，**预期都是替换文件，不会是新增文件。**

2. 替换本地环境的KEY—校验公钥

1. 抓取android branch
2. cd vendor/mediatek/proprietary/scripts/sign-image_v2/der_extractor
3. python pem_to_der.py **root_pubk.pem** **root_pubk.der**
4. chmod 777 der_extractor
5. cd android根目录
6. ./vendor/mediatek/proprietary/scripts/sign-image_v2/der_extractor/der_extractor root_pubk.der **oemkey.h**
ANDROID_SBC

注：

root_pubk.pem 为第4页生成的android root 公钥，该步骤用于生成校验用的oemkey.h文件

2. 准备本地环境的KEY—校验公钥

更新以下文件对应code

Yocto lk2	<yocto code>/src/bsp/lk2/target/auto8668p1_64_ufs/include/oemkey.h
Adnroid lk2	<yocto code>/src/bsp/lk2/target/auto8668p1_64_ufs/include/oemkey.h （最新的版本lk2_an和lk2合并为一包code了）

不要直接替换oemkey.h文件，将新gen出的oemkey.h内的OEM_PUBK覆盖原始oemkey.h的对应数组，不要修改其他配置（如果是RSA3072 key, 请覆盖对应的OEM_3072_PUBK_SZ）

```
#define OEM_PUBK_SZ 256

#define OEM_PUBK
0xDA, 0xCD, 0x8B, 0x5F, 0xDA, 0x8A, 0x76, 0x6F, 0xB7, 0xBC, 0xAA, 0x43, 0xF0, 0xB1, 0x69, 0x15, \
0xCE, 0x7B, 0x47, 0x71, 0x4F, 0x13, 0x95, 0xFD, 0xEB, 0xCF, 0x12, 0xA2, 0xD4, 0x11, 0x55, 0xB0, \
0xFB, 0x58, 0x7A, 0x51, 0xFE, 0xCC, 0xCB, 0x4D, 0xDA, 0x1C, 0x8E, 0x5E, 0xB9, 0xEB, 0x69, 0xB8, \
0x6D, 0xAF, 0x2C, 0x62, 0x0F, 0x6C, 0x27, 0x35, 0x21, 0x5A, 0x5F, 0x22, 0xC0, 0xB6, 0xCE, 0x37, \
0x7A, 0xA0, 0xD0, 0x7E, 0xB3, 0x8E, 0xD3, 0x40, 0xB5, 0x62, 0x9F, 0xC2, 0x89, 0x04, 0x94, 0xB0, \
0x78, 0xA6, 0x3D, 0x6D, 0x07, 0xFD, 0xEA, 0xCD, 0xBE, 0x3E, 0x7F, 0x27, 0xFD, 0xE4, 0xB1, 0x43, \
0xF4, 0x9D, 0xB4, 0x97, 0x14, 0x37, 0xE6, 0xD0, 0x0D, 0x9E, 0x18, 0xB5, 0x6F, 0x02, 0xDA, 0xBE, \
0xB0, 0x00, 0x0B, 0x6E, 0x79, 0x51, 0x6D, 0x0C, 0x80, 0x74, 0xB5, 0xA4, 0x25, 0x69, 0xFD, 0x0D, \
0x91, 0x96, 0x65, 0x5D, 0x2A, 0x40, 0x30, 0xD4, 0x2D, 0xFE, 0x05, 0xE9, 0xF6, 0x48, 0x83, 0xE6, \
0xD5, 0xF7, 0x9A, 0x5B, 0xFA, 0x3E, 0x70, 0x14, 0xC9, 0xA6, 0x28, 0x53, 0xDC, 0x1F, 0x21, 0xD5, \
0xD6, 0x26, 0xF4, 0xD0, 0x84, 0x6D, 0xB1, 0x64, 0x52, 0x18, 0x7D, 0xD7, 0x76, 0xE8, 0x88, 0x6B, \
0x48, 0xC2, 0x10, 0xC9, 0xE2, 0x08, 0x05, 0x9E, 0x7C, 0xAF, 0xC9, 0x97, 0xFD, 0x2C, 0xA2, 0x10, \
0x77, 0x5C, 0x1A, 0x5D, 0x9A, 0xA2, 0x61, 0x25, 0x2F, 0xB9, 0x75, 0x26, 0x8D, 0x97, 0x0C, 0x62, \
0x73, 0x38, 0x71, 0xD5, 0x78, 0x14, 0x09, 0x8A, 0x45, 0x3D, 0xF9, 0x2B, 0xC6, 0xCA, 0x19, 0x02, \
0x5C, 0xD9, 0xD4, 0x30, 0xF0, 0x2E, 0xE4, 0x6F, 0x80, 0xDE, 0x6C, 0x63, 0xEA, 0x80, 0x2B, 0xEF, \
0x90, 0x67, 0x3A, 0xAC, 0x4C, 0x66, 0x67, 0xF2, 0x88, 0x3F, 0xB4, 0x50, 0x1F, 0xA7, 0x74, 0x55
```

3. 配置本地环境

配置<yocto branch> /meta/meta-mediatek-mt8668/conf/machine/auto8668p1_64.conf

```
198 HSM_SUPPORT = "yes"  
199 HSM_SIGN_TOOL = "hsm_sign_tool.py"  
200 HSM_EXEC_CMD = "python3"  
201 #for secure boot dm-verity
```

配置<yocto branch> /meta/meta-mediatek-mt8668-hyp/conf/machine/auto8668p1_64_uos.inc

```
27 HSM_SUPPORT:uos-os = "yes"  
28 HSM_SIGN_TOOL:uos-os = "hsm_sign_tool.py"  
29 HSM_EXEC_CMD:uos-os = "python3"  
30 DM_VERITY_IMAGE_NAME = "hsm_sign_tool.py"
```

3. 配置本地环境

<yocto branch>/meta/meta-mediatek-mt8668/scripts/sign-image_v2/hsm.py

<yocto branch>/meta/meta-mediatek/recipes-bsp/lk/files/pbp/hsm.py

<yocto branch>/meta/meta-mediatek/recipes-bsp/secure-boot/hsm-sign-env/hsmsigntool/hsm_sign_tool.py

<android branch>/vendor/mediatek/proprietary/scripts/sign-image_v2/hsm.py

客制化 hsm_rsa_sign 函数

DA 远程签名

1. 准备公钥和证书

从安全部门获取secure boot 使用的公钥，可以跟之前的key一样（root和img就是前面文档生成的两把key）

Key Name	Key 类型	Key 用途
root_pubk.pem	Android root 公钥	签名auth file
img_pubk.pem	Android image 公钥	签名auth file
da_pubk.pem	Android da 公钥	签名da

2. 替换本地环境的KEY

将上一步的三把公钥改成跟图片高亮文件一样的名字，更新到
<android branch>/
vendor/mediatek/proprietary/scripts/secure_chip_tools/custom_keys
比如da_pubk.pem 改名成da_prvk.pem 覆盖该目录文件。

< vendor > mediatek > proprietary > scripts > secure_chip_tools > custom_keys

Name	Date modified	Type	Size
da_prvk.pem	8/30/2024 9:44 AM	PEM File	2 KB
epp_prvk.pem	8/30/2024 9:44 AM	PEM File	2 KB
img_prvk.pem	8/30/2024 9:44 AM	PEM File	2 KB
img_pubk.pem	8/30/2024 9:44 AM	PEM File	1 KB
primary_dbg_prvk.pem	8/30/2024 9:44 AM	PEM File	2 KB
root_prvk.pem	8/30/2024 9:44 AM	PEM File	2 KB
root_pubk.pem	8/30/2024 9:44 AM	PEM File	1 KB
secondary_dbg_prvk.pem	8/30/2024 9:44 AM	PEM File	2 KB
sla_prvk.pem	8/30/2024 9:44 AM	PEM File	2 KB

3. 配置本地环境

客制化<android branch>/
vendor/mediatek/proprietary/scripts/secure_chip_tools/custom_sign.py 的
hsm_rsa_sign 函数。

```
def hsm_rsa_sign(data, key, padding, sig, oem_args):
```

Data: hash 值

Key: 公钥

Padding: 算法的padding属性

Sig: 签名后的signature指针

Oem_args: 客制化参数

4. 执行签名命令

1. 将da 放到<android branch>/
vendor/mediatek/proprietary/scripts/secure_chip_tools/ 目录下
2. 签名da
python MTK/resign_da.py DA_BR.bin
MT6881 ./settings/Legacy/da/bbchips_pss.ini all DA_BR_resigned.bin
3. 签名auth file
python MTK/toolauth.py -i ./settings/Legacy/authfile/toolauth_key.ini -
g ./settings/Legacy/authfile/toolauth_gfh_config_pss.ini da.auth

原始da: DA_BR.bin

签名后的da: DA_BR_resigned.bin

签名后的auth file: da.auth

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